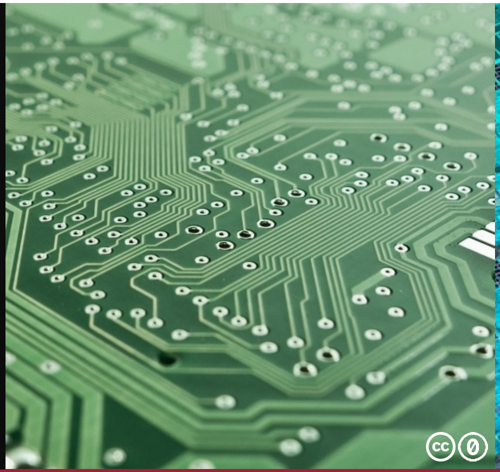
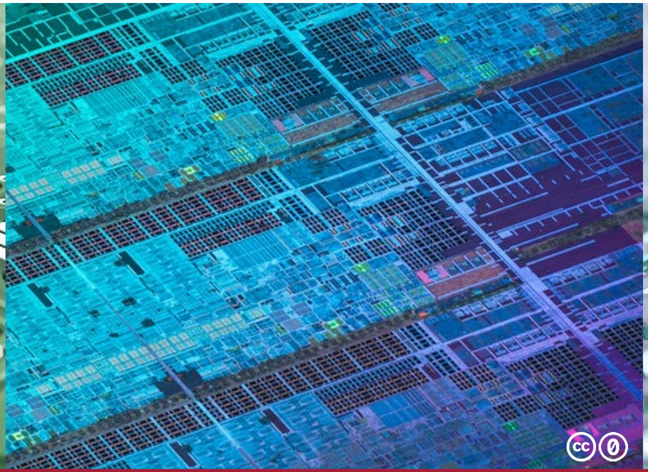
Modern Search Engines


Search Engine Implementation

Group Project

06.08.2026 | Nico Henzler, Lukas Henzler, Tuba Aykan, Samuel Scheer

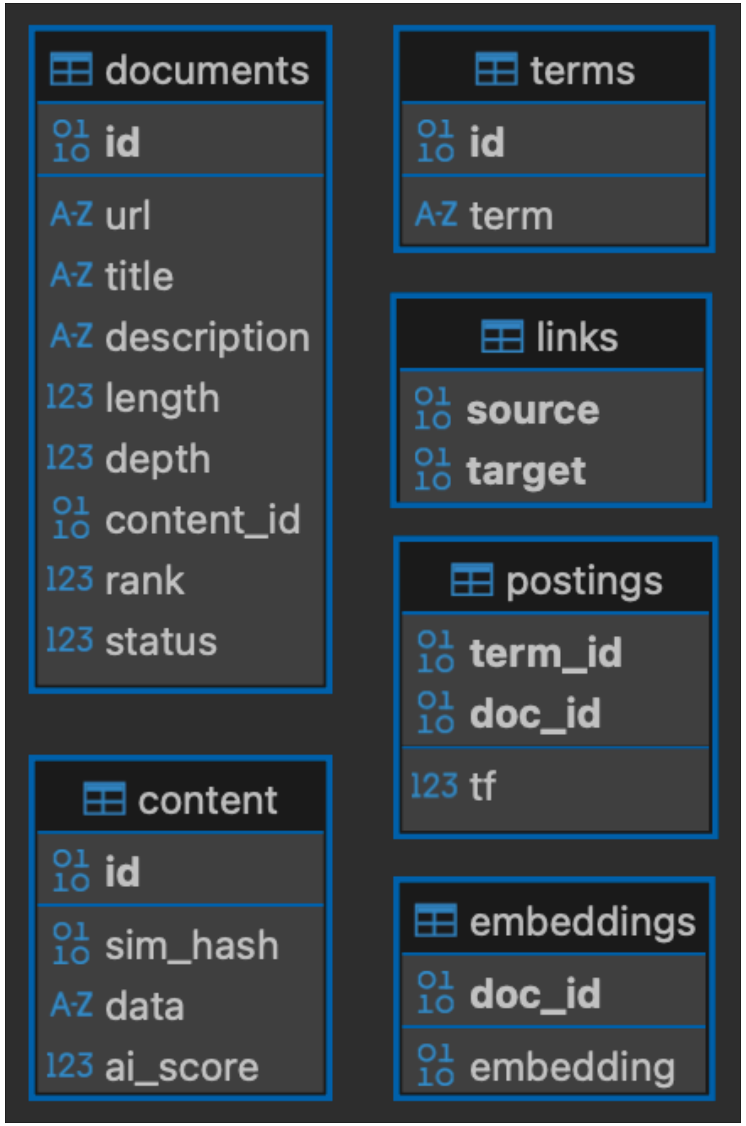
Outline

- Database
- Preprocessing
- Ranking
- Live Demo and Batch Queries
- Summary

Database

- SQLite as source of truth:
  - Document metadata
  - Crawl state
  - Terms and postings

- Storage and efficiency optimisations:
  - **Strict tables**:
    Enforce types for consistency
  - **Deduplication**:
    UNIQUE Key on normalised URLs
  - **SQL aggregation**:
    Count matching docs within DB (TF-IDF)
  - **BLOBs**:
    Store embedding float arrays as binary
  - **Only read what is required**



Preprocessing

Crawler

- configurable threads, default 12
- Caches robots files
- Keeps status to resume after interrupts
  - Pending
  - Cached
  - Skipped
  - Error

PageRank

- Utilises stored link graph after crawling

AI-Detection

- AICodexLab ModernBERT

Indexer

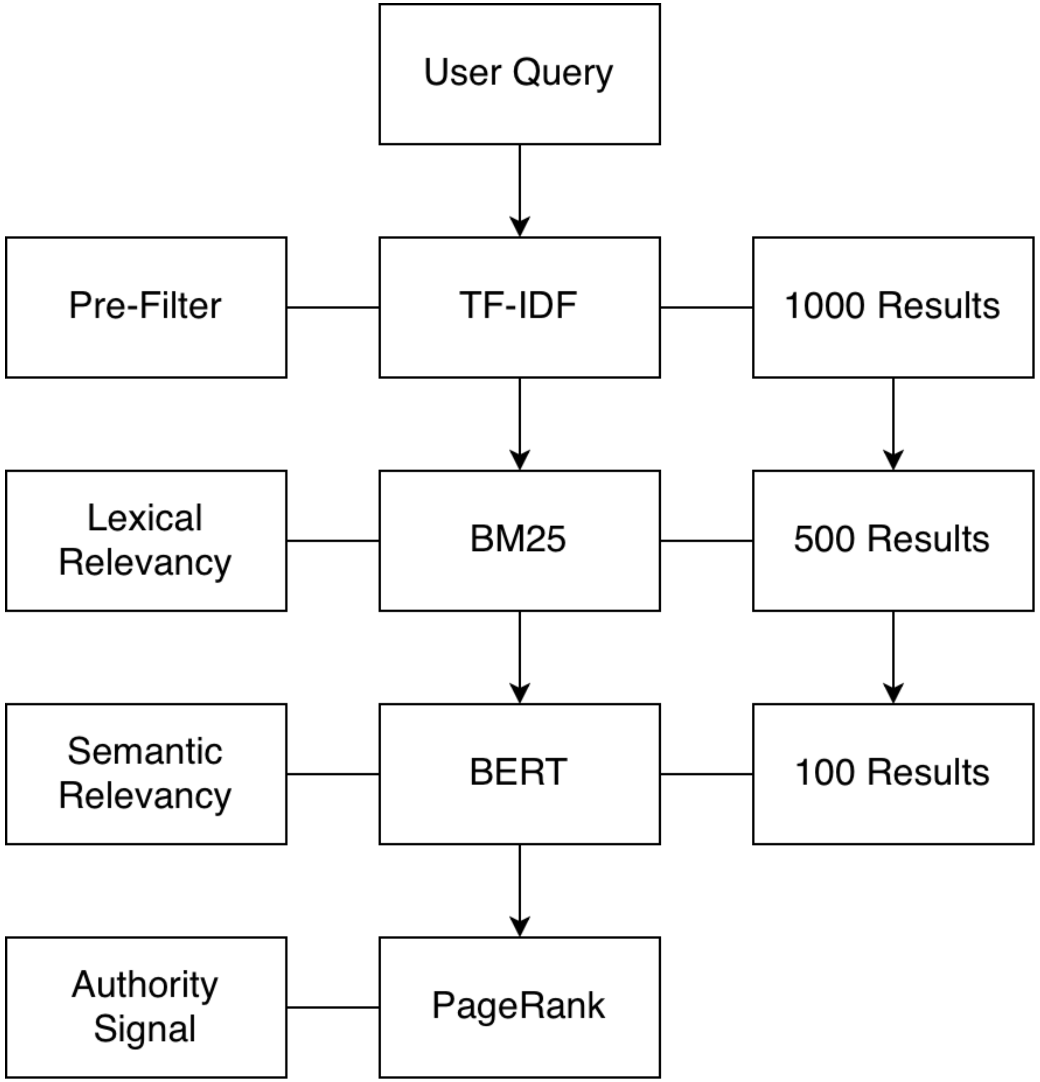
- configurable processes, default 12
- Remove stop words and normalise umlauts
- Stemming was often to aggressive (using Porter2 and Snowball)
- Lemmatisation gave cleaner matching
- 64-bit SimHash: Hamming & Jaccard distance

BERT

- Our own fine-tuned BERT Base Uncased for 10k steps on MS MARCO BM25
- Embeddings: Title, description, and content
- Utilise CUDA → Apple MPS → CPU

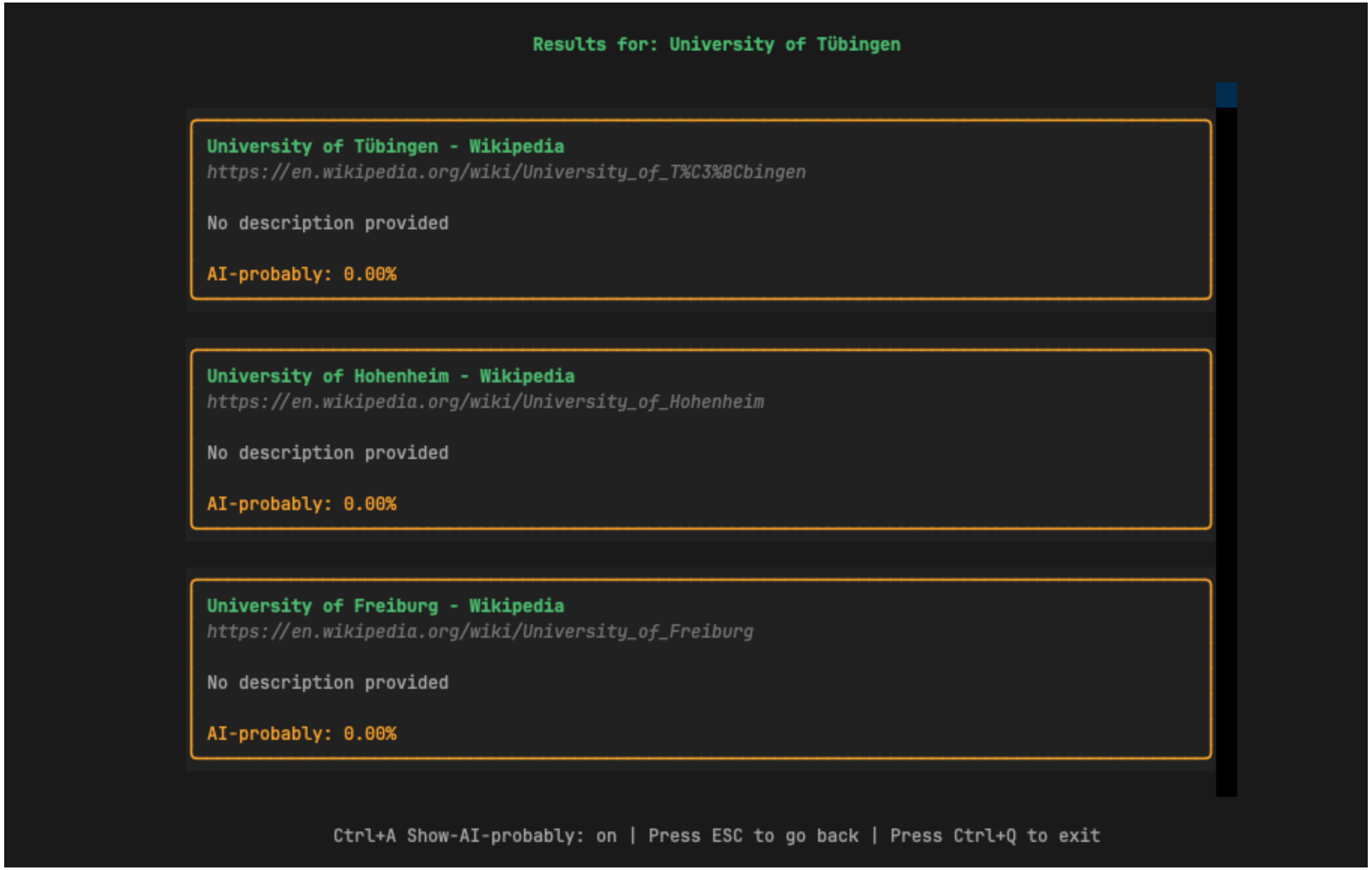
Fast-to-smart Funnel

- **Goal**:
  Spend computation where it matters
- **Lightweight reduction**:
  TF-IDF and BM25
- **Bi-encoder-style**:
  Calculate BERT embeddings separately
- **PageRank**:
  Boost well connected pages
- **Final score**:
  0.7 × BERT + 0.3 × PageRank



Batch Queries &

LIVE DEMO



Summary

- Database as source of truth
- Persistent and restartable pipeline
- Offline pre-computation of PageRank, AI scores, and BERT
- Efficient lexical filtering at query time
- Semantic and authority-aware ranking
- Pure TUI
- Special feature: AI-Detection

QUESTIONS?